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JOINT ARRANGEMENT FOR A ROBOT ARM FOR PROVIDING AN ENERGY-EFFICIENT BRAKING FUNCTION

Abstract

The present invention provides a joint arrangement for a robot arm, especially a robot arm for use in microsurgery, comprising: at least one brake device being configured for engagement and for disengagement and comprising a stator assembly and a rotor assembly, wherein the brake device is further configured to be operable at a minimal power consumption in a disengagement state. The present invention further provides a method for operating a brake device in a joint arrangement for a robot arm, especially a robot arm for use in microsurgery, the method at least comprising the following step: the brake device is provided and configured for engagement and for disengagement, the brake device comprising a stator assembly and a rotor assembly, wherein the brake device is further operated at a minimal power consumption in a disengagement state.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a U.S. National Phase of International Application No. PCT/EP2023/059857 entitled “JOINT ARRANGEMENT FOR A ROBOT ARM FOR PROVIDING AN ENERGY-EFFICIENT BRAKING FUNCTION,” and filed on Apr. 15, 2023. International Application No. PCT/EP2023/059857 claims priority to European Patent Application No. 22169243.7 filed on Apr. 21, 2022. The entire contents of each of the above-listed applications are hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0002] The present invention relates to a joint arrangement for a robot arm having at least one brake device, to a robot arm, and to a method for operating the brake device in the joint arrangement.

BACKGROUND AND SUMMARY

[0003] The present invention particularly relates to a joint arrangement for a robot arm, especially a robot arm for use in microsurgery, having at least one brake device.

[0004] Further, the present invention particularly relates to a robot arm, especially a robot arm for use in microsurgery, comprising said joint arrangement.

[0005] Still further, the present invention particularly relates to a method for operating a brake device in a joint arrangement for a robot arm, especially a robot arm for use in microsurgery.

[0006] Generally, a brake device such as an electro-magnetic brake can be configured such as to provide its braking effect in a currentless state by use of a magnetic field provided by a permanent magnet. In order to neutralize said braking effect, the magnetic field provided by said permanent magnet is usually displaced by an opposing electro-magnetic field, i.e., in a current-carrying state. However, the power needed for remaining in the state of neutralization of said braking effect generally tends to be at a relatively high level leading to undesired heat generation originating from the electro-magnetic brake. In particular, such an electro-magnetic brake is usually operated by the power being ramped up above a neutralization value for neutralizing the braking effect and by the power being ramped down again below a braking effect value for applying the braking effect again, thereby requiring a lot of power. In this respect, the generated heat, i.e., the high temperatures, can consequently transfer to the environment of the brake, e.g., neighboring elements, which can lead to an uncontrolled change of dimensions of said environment/said elements.

[0007] For instance, the above-described effect can be problematic when occurring in applications having the need for high precision such as in surgery procedures, in particular microsurgery or super-microsurgery procedures, such as in joint devices and/or robot devices being used for precise surgical handlings in such procedures.

[0008] Further, in view of economic considerations, the above-described effect can also be considered as disadvantageous (e.g., with respect to high power consumption).

[0009] In view of the above, it is an object of the invention to provide a joint arrangement for a robot arm having at least one brake device, which is capable to be operated in a more economical

and more precise manner.

[0010] According to the invention, this object is obtained by a joint arrangement for a robot arm, especially a robot arm for use in microsurgery, according to claim 1.

[0011] Correspondingly, a joint arrangement for a robot arm, especially a robot arm for use in microsurgery, comprises at least one brake device being configured for engagement and for disengagement and comprising a stator assembly and a rotor assembly, wherein the brake device is further configured to be operable at a minimal power consumption in a disengagement state.

[0012] The at least one brake device may be exactly one brake device or more than one brake device.

[0013] The brake device may be configured as an electro-magnetic brake.

[0014] The minimal power consumption may be a power consumption being necessary for keeping the brake device in a disengagement state, wherein the minimal power consumption may be lower than a power consumption being necessary for switching between an engagement state to said disengagement state.

[0015] In particular, the minimal power consumption may differ from said power consumption being necessary for switching between an engagement state to said disengagement state by at least 0 to 5%, preferably 0 to 10% (e.g., 5 to 10%), more preferably 0 to 15% (e.g., 5 to 15%, or 10 to 15%), even more preferably 0 to 20% (e.g., 5 to 20%, or 10 to 20%, or 15 to 20%), or 0 to 25% (e.g., 5 to 25%, or 10 to 25%, or 15 to 25%, or 20 to 25%).

[0016] The joint arrangement as described is aiming in one possible arrangement for a 75% power reduction between disengaging and keeping the brake disengaged.

[0017] The invention is based on the basic idea that the brake device of the joint arrangement can be operated at a minimal power consumption, to thereby use less power and generate less heat. Accordingly, the joint arrangement can be operated in a more economic manner, wherein, at the same time, it can also be operated in a more accurate manner since thermal expansions are reduced. Operating at the minimal power consumption is possible by making use of a hysteresis effect of the brake device. In particular, the hysteresis effect of the brake device may occur in a range between a power required for disengagement and a power required for engagement. For instance, if the power falls below the power required for disengagement, but stays above the power required for engagement, the brake device can still remain in a stable state of disengagement. Further, since the heat generation can be reduced, the joint arrangement may also be improved with regard to its safety conditions.

[0018] The brake device may comprise at least one permanent magnet for providing a braking effect during engagement and being assigned to the rotor assembly, and at least one electromagnet for releasing the braking effect during disengagement and being assigned to the stator assembly, wherein the permanent magnet and the electromagnet may be arranged or arrangeable to each other with a gap between themselves such as to allow the brake device to be operable at a minimal power consumption. In other words, the permanent magnet and the electromagnet may be arranged or arrangeable to each other with a gap between themselves in order to maintain a hysteresis effect between themselves, when operated.

[0019] Alternatively, the permanent magnet can be replaced by a spring element or any element, that can apply a force for providing a braking effect.

[0020] The gap may be smaller than $1\text{ mm} \pm 10\%$, in particular smaller than $1\text{ mm} \pm 5\%$, particularly smaller than $1\text{ mm} \pm 3\%$. For example, the gap may be in a range between $0.2\text{ mm} \pm 5\%$ and $0.5\text{ mm} \pm 5\%$, or the gap may be in a range between $0.5\text{ mm} \pm 5\%$ and $0.8\text{ mm} \pm 5\%$, or the gap may be in a range between $0.8\text{ mm} \pm 5\%$ and $1\text{ mm} \pm 5\%$.

[0021] Alternatively, the gap may be smaller than $0.8\text{ mm} \pm 10\%$, in particular smaller than $0.8\text{ mm} \pm 5\%$, particularly smaller than $0.8\text{ mm} \pm 3\%$.

[0022] It may also be conceivable that the gap may be smaller than $0.5\text{ mm} \pm 10\%$, in particular smaller than $0.5\text{ mm} \pm 5\%$, particularly smaller than $0.5\text{ mm} \pm 3\%$.

[0023] In particular, the gap may be $0.5\text{ mm}\pm 10\%$, in particular $0.5\text{ mm}\pm 5\%$, particularly $0.5\text{ mm}\pm 3\%$.

[0024] As one possible example, the gap between brake and brake-disc can be $0.2\text{ mm}+0.1\text{ mm}$ and so, it is normally in the range of 0.20 mm to 0.30 mm . The tolerances for the gap can be 0.0171 to 0.329 mm .

[0025] The joint arrangement may further comprise at least one motor stack having a rotor assembly and a stator assembly, wherein the brake device may be arranged in connection with or within the motor stack. In particular, the brake device may be arranged in connection with or at a central motor shaft of the motor stack. In other words, the brake device may be arranged for allowing the brake device to apply a braking effect onto the motor stack, in particular, onto the rotor assembly of the motor stack. Hence, the brake device is configured to apply a braking effect onto the motor stack, in particular, onto the rotor assembly of the motor stack.

[0026] The rotor assembly of the brake device may be configured as a part of the rotor assembly of the motor stack, and the stator assembly of the brake device may be configured as a part of the stator assembly of the motor stack. For example, the rotor assembly of the brake device may be fixed to the rotor assembly of the motor stack by a shaft hub connection, thereby forming a part of the rotor assembly of the motor stack. However, any other suitable connection may also be applicable. Hence, the braking effect applicable by the brake device may be provided to the motor stack in a more direct manner, e.g., without any intermediate element (e.g., transmitting elements).

[0027] The motor stack may comprise at least one spring, in particular at least one leaf spring, being configured to provide a spring force to the brake device. For example, the spring may be arranged in connection with or at a/the central motor shaft of the motor stack, to thereby provide said spring force to the brake device also being arranged in connection with or at said central motor shaft. In other words, the spring force may be provided to the brake device via the central motor shaft. In particular, the spring may be configured to provide a/the spring force to the brake device for intensifying a/the hysteresis effect of the brake device.

[0028] Further, according to the invention, the above-described object is also obtained by a robot arm, especially a robot arm for use in microsurgery, according to claim 7.

[0029] Correspondingly, a robot arm, especially a robot arm for use in microsurgery, comprises a joint arrangement, for example, a/the joint arrangement as described herein, e.g., as described above and/or further below.

[0030] The robot arm may further comprise one or more further joint arrangements being particularly configured as described herein, e.g., as described above and/or further below.

[0031] In other words, the robot arm may comprise one or more joint arrangements, wherein at least one joint arrangement may be configured as described herein, e.g., as described above and/or further below.

[0032] Still further, according to the invention, the above-described object is also obtained by a method for operating a brake device in a joint arrangement for a robot arm, especially a robot arm for use in microsurgery, according to claim 8.

[0033] Correspondingly, a method for operating a brake device in a joint arrangement (e.g., a/the joint arrangement as described herein, e.g., as described above and/or further below) for a robot arm (e.g., a/the robot arm as described herein, e.g., as described above and/or further below), especially a robot arm for use in microsurgery, the method at least comprising the following step: the brake device is provided and configured for engagement and for disengagement, the brake device comprising a stator assembly and a rotor assembly, wherein the brake device is further operated at a minimal power consumption in a disengagement state.

[0034] Accordingly, as described above, due to the method, the brake device of the joint arrangement can be operated at a minimal power consumption, to thereby use less power and generate less heat. Hence, the joint arrangement can be operated in a more economic manner, wherein, at the same time, it can also be operated in a more accurate manner since thermal

expansions are reduced.

[0035] The brake device may further be configured for engagement at an engagement threshold value and for disengagement at a disengagement threshold value being higher than the engagement threshold value. Hence, the brake device may reliably function as a holding brake. In case of a loss of power, the joint arrangement may be prevented to continue moving by the brake device, since the engagement threshold value may be smaller than the disengagement threshold value.

[0036] The method may further comprise the following steps: providing full power to the brake device by applying Pulse Width Modulation (PWM) for disengaging the brake device: reducing the power provided to the brake device to a first predetermined power value by applying PWM, wherein the first predetermined power value is set between the disengagement threshold value and the engagement threshold value: and maintaining the power provided to the brake device at the first predetermined power value by applying PWM for keeping the brake device in disengagement. Hence, the brake device may be switched to its disengagement state by the application of a full power PWM signal and, subsequently, may be maintained therein due to the application of the reduced power PWM signal being lower than the disengagement threshold value and higher than the engagement threshold value, thereby making use of the hysteresis of the brake device and, accordingly, saving power and generating less heat.

[0037] The method may further comprise reducing the power provided to the brake device to a second predetermined power value by applying PWM for engaging the brake device, wherein the second predetermined power value may be set below the engagement threshold value.

[0038] The first predetermined power value may be half of the full power being providable or provided to the brake device by applying pulse width modulation (PWM) for disengaging the brake device. Hence, due to halving the power, an energy reduction of approximately 25% may be achieved or achievable.

[0039] Additionally and/or alternatively, by halving the voltage a power reduction of 75% can be achieved.

[0040] The power reduction can also be influenced by PWM, where the PWM equivalent of halving the voltage is applied giving a power reduction of approx. 75%.

[0041] The second predetermined power value may be zero, i.e., substantially no power.

[0042] The full power being providable or provided to the brake device by applying PWM for disengaging the brake device may include a 24V signal. In this respect, the first predetermined power value may be 12V. However, any other signal may also be conceivable with respect to the full power, such as a 12V signal or a 48V signal. The second predetermined power value may be 0V (i.e., zero).

[0043] The disengagement threshold value may be $\frac{5}{8}$ of the full power. For example, in case of the full power including a 24V signal, the disengagement threshold value may be 15V.

[0044] The engagement threshold value may be $\frac{1}{3}$ of the full power. For example, in case of the full power including a 24V signal, the engagement threshold value may be 8V.

[0045] The step of providing full power to the brake device by applying PWM for disengaging the brake device may be maintained for a first predetermined time period, and the step of maintaining the power provided to the brake device at the first predetermined power value by applying PWM for keeping the brake device in disengagement may be maintained for a second predetermined time period, wherein the second predetermined time period may be longer than the first predetermined time period. Hence, full power may (e.g., only) be used for switching the brake device into the disengagement state for a relatively short period of time (e.g., compared to maintaining the disengagement state), thereby ensuring that the brake device may be operated in an economic way as well as that excessive heat generation may be prevented.

[0046] The brake device may comprise at least one permanent magnet for providing a braking effect during engagement and at least one electromagnet for releasing the braking effect during disengagement. The electromagnet may be configured to provide an electro-magnetic field for

opposing the magnetic field of the permanent magnet for neutralization of the braking effect of the brake device.

[0047] The brake device may be arranged in connection with or within a motor stack of the joint arrangement.

[0048] The motor stack may comprise at least one spring, in particular at least one leaf spring, being configured to provide a spring force to the brake device.

[0049] It should be understood that any feature described in connection with the joint arrangement and/or with the robot arm may also be part of the method for operating a brake device in a joint arrangement, and vice versa. It should be further understood that any advantage and/or property described in connection with the joint arrangement and/or with the robot arm may also apply to the method for operating a brake device in a joint arrangement, and vice versa.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0050] Further details and advantages of the present invention shall now be disclosed in connection with the drawings, where:

[0051] FIG. **1** schematically shows a cross sectional view of a joint arrangement for a robot arm according to an exemplary embodiment of the invention;

[0052] FIG. **2A** schematically shows a cross sectional view of a motor stack of a joint arrangement for a robot arm according to another exemplary embodiment of the invention;

[0053] FIG. **2B** schematically shows a cross sectional view of a part of the motor stack of FIG. **2A**;

[0054] FIG. **3** schematically shows a robot arm according to another exemplary embodiment of the invention;

[0055] FIG. **4** shows a diagram depicting a method for operating a brake device in a joint arrangement for a robot arm according to another exemplary embodiment of the invention;

[0056] FIG. **5** schematically shows a diagram depicting an operating behavior of the method of FIG. **4**;

[0057] FIG. **6** shows a diagram depicting an assembly method of the motor stack of FIG. **2A** according to another exemplary embodiment of the invention;

[0058] FIG. **7** schematically shows a part of the assembly method of FIG. **6**;

[0059] FIG. **8** schematically shows a part of the assembly method of FIG. **6**;

[0060] FIG. **9** schematically shows a part of the assembly method of FIG. **6**;

[0061] FIG. **10** schematically shows a part of the assembly method of FIG. **6**;

[0062] FIG. **11** schematically shows a part of the assembly method of FIG. **6**;

[0063] FIG. **12A-12C** schematically show alternatives of a part of the assembly method of FIG. **6**, respectively;

[0064] FIG. **13** schematically shows a part of the assembly method of FIG. **6**; and

[0065] FIG. **14** schematically shows a part of the assembly method of FIG. **6**.

DETAILED DESCRIPTION

[0066] Identical or functionally equivalent elements are denoted with the same reference signs throughout the figures.

[0067] Turning to FIG. **1**, a cross sectional view of a joint arrangement **10** for a robot arm **20** according to an exemplary embodiment of the invention is schematically shown.

[0068] The joint arrangement **10** comprises a brake device **100**.

[0069] The brake device **100** is configured as an electro-magnetic brake.

[0070] The brake device **100** is configured for engagement and for disengagement.

[0071] Engagement corresponds to a state, in which a braking effect is applied by the brake device **100**, and disengagement corresponds to a state, in which the breaking effect is not applied by the

brake device **100**.

[0072] The brake device **100** is further configured to be operable at a minimal power consumption in a/the disengagement state, which will be described further below.

[0073] The brake device **100** comprises a stator assembly **102** and a rotor assembly **104**.

[0074] The brake device **100** further comprises a permanent magnet **106** for providing a braking effect during engagement.

[0075] The permanent magnet **106** is assigned to the rotor assembly **104**.

[0076] The permanent magnet **106** is constructed to be substantially axially symmetric relative to a central axis.

[0077] The brake device **100** also comprises an electromagnet **108** for releasing the braking effect during disengagement.

[0078] The electromagnet **108** is assigned to the stator assembly **102**.

[0079] The electromagnet **108** is constructed to be substantially axially symmetric relative to a central axis.

[0080] The permanent magnet **106** and the electromagnet **108** are disposed opposite to each other in such a way that their symmetric axes coincide. In this respect, the permanent magnet **106** and the electromagnet **108** are arranged or arrangeable to each other with a gap G between themselves.

[0081] The gap G between the permanent magnet **106** and the electromagnet **108** is dimensioned such as to allow the brake device **100** to be operable at a minimal power consumption, e.g., to be allowed to make use of a hysteresis effect between themselves, when operated, in order to be operable at a minimal power consumption.

[0082] In the present embodiment, the gap G is $0.5\text{ mm}\pm 10\%$.

[0083] However, it may also be conceivable that the gap G is alternatively dimensioned such as the following: the gap may be smaller than $1\text{ mm}\pm 10\%$, in particular smaller than $1\text{ mm}\pm 5\%$, particularly smaller than $1\text{ mm}\pm 3\%$. For example, the gap may be in a range between $0.2\text{ mm}\pm 5\%$ and $0.5\text{ mm}\pm 5\%$, or the gap may be in a range between $0.5\text{ mm}\pm 5\%$ and $0.8\text{ mm}\pm 5\%$, or the gap may be in a range between $0.8\text{ mm}\pm 5\%$ and $1\text{ mm}\pm 5\%$. Alternatively, the gap may be smaller than $0.8\text{ mm}\pm 10\%$, in particular smaller than $0.8\text{ mm}\pm 5\%$, particularly smaller than $0.8\text{ mm}\pm 3\%$. It may also be conceivable that the gap may be smaller than $0.5\text{ mm}\pm 10\%$, in particular smaller than $0.5\text{ mm}\pm 5\%$, particularly smaller than $0.5\text{ mm}\pm 3\%$.

[0084] As can be seen in FIG. 1, the joint arrangement **10** further comprises a motor stack **110**.

[0085] The motor stack **110** is disposed in the joint arrangement **10** for driving a pulley **112** via a belt **114** for transmitting a torque.

[0086] Further, in the exemplary embodiment, the joint arrangement **10** also comprises a control unit **116** for controlling the motor stack **110** and/or the brake device **100**.

[0087] However, it may also be conceivable that the control unit **116** may be configured as an element being separate from the joint arrangement **10** and having a connection with the joint arrangement **10** for signal transmission.

[0088] The motor stack **110** comprises a rotor assembly **118** and a stator assembly **120**.

[0089] The rotor assembly **118** of the motor stack **110** is partly formed by a central motor shaft **122** of the motor stack **10**.

[0090] The brake device **100** is arranged in connection with or within the motor stack **110**.

[0091] In particular, the brake device **100** is arranged at said central motor shaft **122** of the motor stack **110**.

[0092] Hence, the brake device **100** is arranged for allowing the brake device **100** to apply a braking effect onto the motor stack **110**, in particular, onto the rotor assembly **118** of the motor stack **100**.

[0093] In this respect, the rotor assembly **104** of the brake device **100** is rotatable together with the rotor assembly **118** of the motor stack **110**, and the stator assembly **102** of the brake device **100** is non-rotatable, e.g., fixed, together with the stator assembly **120** of the motor stack **110**.

[0094] Therefore, the rotor assembly **104** of the brake device **100** can be seen as being configured as a part of the rotor assembly **118** of the motor stack **110**, and the stator assembly **102** of the brake device **100** can be seen as being configured as a part of the stator assembly **120** of the motor stack **110**.

[0095] For example, in the present embodiment, the rotor assembly **104** of the brake device **100** is fixed to the rotor assembly **118** of the motor stack **110** by a shaft hub connection **124**, thereby forming a/the part of the rotor assembly **118** of the motor stack **110**.

[0096] Accordingly, the rotor assembly **118** of the motor stack **110** and the rotor assembly **104** of the brake device **100** are stopped or stoppable together in case of engagement of the brake device **100** and are released or releasable again together in case of disengagement of the brake device **100**.

[0097] Now turning to FIGS. 2A and 2B, a cross sectional view of a motor stack **110** of a joint arrangement **10** for a robot arm **20** according to another exemplary embodiment of the invention is schematically shown in more detail.

[0098] The motor stack **110** shown in FIGS. 2A and 2B is substantially the same as described above, in particular, with respect to the joint arrangement **10** shown in FIG. 1, so that only the differences shall be described in the following.

[0099] It should be understood that the motor stack **110** of FIG. 1 may be substituted with the motor stack **110** of FIGS. 2A and 2B. Hence, the joint arrangement **10** of FIG. 1 may alternatively comprise the motor stack **110** of FIG. 2A and 2B.

[0100] In other words, the motor stack **110** of FIGS. 2A and 2B may be configured for integration into a joint arrangement **10** for a robot arm **20**, in particular, in the joint arrangement **10** of FIG. 1.

[0101] Referring to FIG. 2A, the motor stack **110** further comprises a first housing **126**, a second housing **128**, an intermediate flange **130** and a cover **132**.

[0102] The first housing **126** is connected or connectable to the second housing **128**, and the second housing **128** is connected or connectable to the intermediate flange **132** and the cover **132**.

[0103] A respective sealing is disposed between the first housing **126** and the second housing **128**, and between the second housing **128** and the cover **132**.

[0104] The motor shaft **122** is substantially enclosed by the cover **130** and the first and the second housing **126**, **128** and protrudes through the first housing **126** for being able to output a torque.

[0105] The motor stack **110** further comprises a first bearing **134** and a second bearing **136**.

[0106] The first and second bearings **134**, **136** are configured as ball bearings, respectively.

However, any other bearing configuration may also be conceivable.

[0107] The first bearing **134** is disposed in the first housing **126** for rotatably holding the motor shaft **122** at a first position.

[0108] In particular, the first bearing **134** is disposed in an/the opening of the first housing **126**, through which the motor shaft **122** extends for protruding from the first housing **126** in order to output a/the torque.

[0109] The brake device **100** is arranged in the first housing **126** and adjacent to the first bearing **134**.

[0110] In particular, the electromagnet **108** of the brake device **100** is fixed to the first housing **126**.

[0111] The motor shaft **122** extends through a central passage formed in the electromagnet **108** such as to be able to rotate.

[0112] The permanent magnet **106** is fixed to a flange **138** of the motor stack **110**, i.e., the motor stack **110** further comprises a flange **138** for connecting to the brake device **100**, in particular, to the permanent magnet **106**. However, it may also be conceivable that the flange **138** may be configured as a part of the brake device **100**.

[0113] Said flange **138** (together with the permanent magnet **106**) is fixed to the motor shaft **122** by the before-mentioned shaft hub connection **124**.

[0114] In this respect, the permanent magnet **106** (on the flange **138**) and the electromagnet **108** are arranged to form the above-described gap **G** between themselves.

[0115] The second bearing **136** is disposed in the intermediate flange **130** being fixed to the second housing **128** for rotatably holding the motor shaft **122** at a second position.

[0116] Hence, the second housing **128** is fixed to the first housing **126** at one end thereof and to the cover **132** at the opposite end thereof, wherein the intermediate flange **130** and, accordingly, the second bearing **136** are arranged on said opposite end between the cover **132** and the second housing **128**.

[0117] As can be seen in FIG. 2B, the intermediate flange **130** is pretensioned against the second housing **128** by at least one pretension spring **148**, for example, three pretension springs **148** being equally arrangeable at the intermediate flange **130** with respect to (e.g., around) a central axis or, in a mounted state, the motor shaft **122**.

[0118] The pretension spring **148** is configured as a compression spring.

[0119] Since the intermediate flange **130** is pretensioned by the pretension spring **148**, the second bearing **136** is also pretensioned, thereby eliminating axial backlash.

[0120] Additionally, the motor stack **110** comprises at least one further spring **150**, in particular at least one leaf spring **150**.

[0121] The leaf spring **150** is fixed to the second housing **128** on one side thereof and in communication with the motor shaft **122** and/or the intermediate flange **130** and/or the rotatable part of the second bearing **136** on the other side thereof.

[0122] The leaf spring **150** is configured to provide a spring force to the brake device **100**, in particular to the permanent magnet **106** of the brake device **100** via the motor shaft **122**. Hence, due to said spring force provided by the leaf spring **150**, a/the hysteresis effect of the brake device **100** can be intensified.

[0123] Further, the leaf spring **150** can support the second bearing in a radial direction (via the motor shaft **122**).

[0124] A driving assembly **140** is arranged in the first and the second housing **126**, **128** and adjacent to the second bearing **136**.

[0125] In other words, the motor stack **110** comprises a driving assembly **140** for driving, e.g., for generating torque.

[0126] The driving assembly **140** (e.g., a motor rotor **140-1**, a motor stator **140-2**, and a motor PCB) is partly arranged at the motor shaft **122** in a rotatable manner (i.e., rotatable part **140-1** of the driving assembly **140**) and at the first and/or the second housing **126**, **128** in a non-rotatable manner (i.e., non-rotatable part **140-2** of the driving assembly **140**) as generally known in the art.

[0127] The driving assembly **140** is connected to a drive PCB **142**, which is arranged in the cover **132** and held therein by a PCB holding structure **144**.

[0128] The drive PCB **142** is signally connected to an encoder **146** disposed at an end of the motor shaft **122**.

[0129] Hence, the motor stack **110** also comprises said drive PCB **142** for controlling the driving assembly **140** as generally known in the art, said PCB holding structure **144**, and said encoder **146** for determining the angular position of the motor shaft **122** as generally known in the art.

[0130] Accordingly, with reference to FIG. 2A, when seen along the motor shaft **122**, the above-described elements are substantially disposed in an order of the first bearing **134**, the electromagnet **108**, the permanent magnet **106**, the flange **138**, the rotatable part **140-1** of the driving assembly **140**, the second bearing **136**, and the encoder **146**.

[0131] It should be understood that additional elements may be disposed on or in connection with the motor shaft **122** such as one or more bushings, one or more retaining rings, one or more sleeves, one or more spacers, one or more retaining washers and/or one or more slotted/grooved nuts (e.g., as exemplarily shown in FIG. 2A).

[0132] Examples of such additional elements are exemplarily shown and exemplarily described in connection with a/the assembly method of FIGS. 6 to 14 further below.

[0133] With reference to FIGS. 2A and 2B, the brake device **100** can provide a/the braking effect to

the motor shaft **122** by the permanent magnet **106** (during engagement) and can release the braking effect therefrom by the electromagnet **108** (during disengagement).

[0134] Now turning to FIG. 3, a robot arm **20** according to another exemplary embodiment of the invention is schematically shown.

[0135] The robot arm **20** is particularly configured for use in microsurgery.

[0136] The robot arm **20** comprises a joint arrangement **10**, for example, a/the joint arrangement **10** as described herein, e.g., as described above and/or further below.

[0137] The joint arrangement **10** forms a sub arm **200** of the robot arm **10**.

[0138] The robot arm **10** further comprises further sub arms **202** for articulation purposes.

[0139] The respective sub arms **202** may each comprise at least one brake device **100** and/or at least one motor stack **110** as described herein. Hence, the sub arms **202** may also be formed by one or more further joint arrangements being particularly configured as described herein, e.g., as described above and/or further below.

[0140] In other words, the robot arm **20** may comprises one or more joint arrangements **10**, wherein at least one joint arrangement **10** may be configured as described herein, e.g., as described above and/or further below.

[0141] As can be seen in FIG. 3, the robot arm **20** additionally comprises an adapter assembly **204** for holding a surgical instrument **206**, to thereby be usable in surgery, in particular microsurgery.

[0142] Said adapter assembly **204** is rotatably connected to the joint arrangement **10**.

[0143] Said adapter assembly **204** is rotatable by applying a torque provided or providable by the motor stack **110** in the joint arrangement **10**, wherein the torque is transferred or transferable to the adapter assembly **204** via the belt **114** and the pulley **112** in the joint arrangement **10**.

[0144] Further, the rotation of the adapter assembly **205** can be stopped by operating the brake device **100** assigned to said motor stack **110**.

[0145] Turning to FIGS. 4 and 5, a diagram depicting a method for operating a brake device **100** in a joint arrangement **10** (e.g., the joint arrangement **10** as described in connection with FIGS. 1 to 2B) for a robot arm **20** (e.g., the robot arm **20** as described in connection with FIG. 3) according to another exemplary embodiment of the invention is shown in FIG. 4, and a diagram depicting an operating behavior of the method of FIG. 4 is schematically shown in FIG. 5.

[0146] The method for operating a/the brake device **100** in a/the joint arrangement **10** for a/the robot arm at least comprises the following step: the brake device **100** is provided, wherein the brake device **100** is configured for engagement and for disengagement and comprises a stator assembly **102** and a rotor assembly **104**, wherein the brake device **100** is further operated at a minimal power consumption in a disengagement state.

[0147] Accordingly, due to the method, the brake device **100** of the joint arrangement **10** can be operated at a minimal power consumption, to thereby use less power and generate less heat. Hence, the joint arrangement **10** can be operated in a more economic manner, wherein, at the same time, it can also be operated in a more accurate manner since thermal expansions are reduced.

[0148] The brake device **100** is configured to engage at an engagement threshold value.

[0149] Further, the brake device **100** is configured to disengage at a disengagement threshold value.

[0150] The disengagement threshold value is higher than the engagement threshold value.

[0151] The disengagement threshold value and the engagement threshold value respectively are smaller than a full power provided or providable for operating the brake device **100** (e.g., as intended and/or without misuse).

[0152] Hence, the brake device **100** can reliably function as a holding brake. In case of a loss of power, the joint arrangement **10** can be prevented to continue moving by the brake device **100**, since the engagement threshold value is smaller than the disengagement threshold value.

[0153] In particular, with reference to FIG. 4, the method comprises the step of providing (S1) full power to the brake device **100** by applying Pulse Width Modulation (PWM) for disengaging the

brake device **100**.

[0154] In the present embodiment, said full power being providable or provided to the brake device **100** includes a 24V signal (see PWM signal in FIG. 5). However, any other signal may also be conceivable with respect to the full power, such as a 12V signal or a 48V signal.

[0155] The disengagement threshold value is $\frac{5}{8}$ of the full power.

[0156] Hence, the full power including a 24V signal in the present embodiment, the disengagement threshold value is 15V (see disengagement threshold value in FIG. 5).

[0157] The engagement threshold value is $\frac{1}{3}$ of the full power.

[0158] Hence, the full power including a 24V signal in the present embodiment, the engagement threshold value is 8V (see engagement threshold value in FIG. 5).

[0159] After the disengagement of the brake device **100** due to providing full power thereto, e.g., after the brake device **100** is switched to its (e.g., stable) disengagement state by providing full power thereto, the power is reduced to a first predetermined power value by applying PWM, step S2.

[0160] As can be seen in FIG. 5, the first predetermined power value is set between the disengagement threshold value and the engagement threshold value.

[0161] In other words, the step of reducing (S2) the power provided to the brake device **100** to a/the first predetermined power value by applying PWM is applied, wherein the first predetermined power value is set between the disengagement threshold value and the engagement threshold value.

[0162] The first predetermined power value is half of the full power being providable or provided to the brake device **100** by applying PWM for disengaging the brake device **100**. In this respect, the first predetermined power value is 12V in the present embodiment (see PWM signal in FIG. 5).

[0163] Hence, due to halving the power, an energy reduction of approximately 25% may be achieved or achievable in the present embodiment.

[0164] Then, the power is maintained at the first predetermined power value by applying PWM for keeping the brake device **100** in disengagement, i.e., its (stable) disengagement state, step S3.

[0165] In other words, the step of maintaining (S3) the power provided to the brake device **100** at the first predetermined power value by applying PWM is applied for keeping the brake device **100** in disengagement.

[0166] Hence, the brake device **100** is switched or switchable to its disengagement state by the application of a/the full power PWM signal and, subsequently, is maintained or maintainable therein due to the application of the reduced power PWM signal being lower than the disengagement threshold value and higher than the engagement threshold value, thereby making use of the hysteresis of the brake device **100** and, accordingly, saving power and generating less heat.

[0167] Since the brake device **100** can be kept in disengagement, i.e., its (stable) disengagement state, by applying the PWM signal set below the disengagement threshold value due to exploitation of the hysteresis effect, the brake device **100** can be operated at said minimal power consumption in the disengagement state.

[0168] Accordingly, the brake device **100** of the joint arrangement **10** can be operated using less power and, therefore, generate less heat. Accordingly, the joint arrangement **10** can be operated in a more economic manner, wherein, at the same time, it can also be operated in a more accurate manner since thermal expansions are reduced. Further, since the heat generation can be reduced, the joint arrangement **10** can also be improved with regard to its safety conditions.

[0169] The above-described step of providing (S1) full power to the brake device **100** by applying PWM for disengaging the brake device **100** is maintained for a first predetermined time period.

[0170] Further, the above-described step of maintaining (S3) the power provided to the brake device **100** at the first predetermined power value by applying PWM for keeping the brake device **100** in disengagement is maintained for a second predetermined time period.

[0171] As can be seen in FIG. 5 as being schematically depicted, the second predetermined time

period. is (e.g., much) longer than the first predetermined time period.

[0172] Hence, full power is (e.g., only) used for switching the brake device **100** into the disengagement state for a relatively short period of time (e.g., compared to maintaining the disengagement state), thereby ensuring that the brake device **100** is operated in an economic way as well as that excessive heat generation is prevented.

[0173] In case of the brake device **100** to be switched back to engagement, i.e., its engagement state, the power is reduced again to a second predetermined power value by applying PWM, step **S4**.

[0174] The second predetermined power value is set below the engagement threshold value, i.e., below 8V in the present embodiment. As can be seen in FIG. 5, the second predetermined power value is 0V in the present embodiment (see PWM signal in FIG. 5).

[0175] In other words, the method further comprises the step of reducing (**S4**) the power provided to the brake device **100** to a/the second predetermined power value by applying PWM for engaging the brake device **100**, wherein the second predetermined power value is set below the engagement threshold value.

[0176] It should be understood that any feature described in connection with the joint arrangement **10** and/or with the robot arm **20** may also be part of the method for operating a brake device **100** in a joint arrangement **10**, and vice versa. It should be further understood that any advantage and/or property described in connection with the joint arrangement **10** and/or with the robot arm **20** may also apply to the method for operating a brake device **100** in a joint arrangement **10**, and vice versa.

[0177] In this respect, for example, the brake device **100** comprises at least one permanent magnet **106** for providing a braking effect during engagement and at least one electromagnet **108** for releasing the braking effect during disengagement.

[0178] Accordingly, in another example, the electromagnet **108** is configured to provide an electromagnetic field for opposing the magnetic field of the permanent magnet **106** for neutralization of the braking effect of the brake device **100**.

[0179] Accordingly, in another example, the brake device **100** is arranged in connection with or within a motor stack **110** of the joint arrangement **10**.

[0180] Accordingly, in a still further example, the motor stack **110** comprises at least one spring **150**, in particular at least one leaf spring **150**, being configured to provide a spring force to the brake device **100**.

[0181] Turning to FIG. 6, the present invention additionally provides an assembly method of a/the motor stack **110** as described herein in order to help understanding the invention, in particular, the claimed invention, holistically. Hence, FIG. 6 shows a diagram depicting the assembly method of the motor stack **110** of FIG. 2A according to another exemplary embodiment of the invention.

[0182] It should be understood that any feature described in connection with the joint arrangement **10** and/or with the robot arm **20** may also be part of said assembly method, and vice versa. It should be further understood that any advantage and/or property described in connection with the joint arrangement **10** and/or with the robot arm **20** may also apply to said assembly method, and vice versa.

[0183] In a, e.g., a first, step **S10**, the flange **138** is assembled with the permanent magnet **106**.

[0184] In other words, the assembly method comprises the step of assembling (**S10**) a flange sub assembly.

[0185] The flange sub assembly is substantially formed by the flange **138** and the permanent magnet **106** (see FIG. 7).

[0186] In a further, e.g., a second, step **S20**, the intermediate flange **130** is assembled with the second bearing **136** and the leaf spring **150**.

[0187] In other words, the assembly method further comprises the step of assembling (**S20**) an intermediate flange sub assembly.

[0188] The intermediate flange sub assembly is substantially formed by the intermediate flange

130, the second bearing **136** and the leaf spring **150** (see FIG. **8**).

[0189] In a further, e.g., a third, step **S30**, the first housing **126** is assembled with a plurality of elements of the motor stack **110** as described with respect to FIG. **2A**, such as the flange **138** and the permanent magnet **106**, i.e., the before-mentioned flange sub assembly, thereby substantially forming a first housing sub assembly.

[0190] In other words, the assembly method further comprises the step of assembling (**S30**) a/the first housing sub assembly (see FIGS. **9** to **12C**).

[0191] In a further, e.g., a fourth, step **S40**, the second housing **128** is mounted to the first housing **126**, i.e., to the first housing sub assembly.

[0192] In other words, the assembly method further comprises the step of mounting (**S40**) the second housing **128** to a/the first housing sub assembly (see FIG. **13**).

[0193] In a further, e.g., a fifth, step **S50**, the intermediate flange sub assembly is mounted to the second housing **128** and to the first housing sub assembly having the second housing **128** mounted thereto.

[0194] In other words, the assembly method further comprises the step of mounting (**S50**) the intermediate flange sub assembly to the second housing **128** and to the first housing sub assembly (see FIG. **14**).

[0195] In a further, e.g., a sixth, step **S60**, the cover **132** is assembled with a plurality of elements of the motor stack **110** as described with respect to FIG. **2A**, thereby substantially forming a cover sub assembly, and then the cover sub assembly is mounted to the assembly resulting from the (e.g., fifth) step **S50**, i.e., substantially formed by the intermediate flange sub assembly, the second housing **128** and the first housing sub assembly (not particularly shown in the figures).

[0196] Accordingly, a/the motor stack **110**, e.g., the motor stack **110** of FIG. **2A**, can be assembled by performing steps **S10** to **S60**.

[0197] It should be understood that the above-described order of steps may be changed with respect to technical reasonability. For example, the first step **S10** and the second step **S20** may be interchanged and/or may be applied during the third step **S30** and/or the second step **S20** may be applied after the fourth step **S40**, and so on. It should be further understood that any of the above-described steps may also be applied in a parallel manner with respect to technical reasonability.

[0198] Turning to FIG. **7**, a part of the assembly method of FIG. **6** is schematically shown.

[0199] In particular, the flange sub assembly of step **S10** is schematically shown.

[0200] The flange sub assembly comprises the flange **138**, a brake armature **300** of the brake device **100** having the permanent magnet **106** (e.g., at least one permanent magnet **106**) arranged therein or thereto, and at least one fastener **302**.

[0201] The fastener **302** is arranged to fix the brake armature **300** having the permanent magnet **106** to the flange **138**.

[0202] In other words, the permanent magnet **106** is fixed to the flange **138** by means of at least one fastener **302**.

[0203] The fastener **302** is configured as a screw fastener.

[0204] Turning to FIG. **8**, a part of the assembly method of FIG. **6** is schematically shown.

[0205] In particular, the intermediate flange sub assembly of step **S20** is schematically shown.

[0206] The intermediate flange sub assembly comprises the intermediate flange **130**, the second bearing **136** and the leaf spring **150**.

[0207] As can be seen in FIG. **8**, the second bearing **136** is pressed into the intermediate flange **130** by means of a first tool **400**, in particular, a press tool.

[0208] While pressing the second bearing **136** into the intermediate flange **130**, the intermediate flange **130** is countered into the pressing direction by a second tool **402**, in particular, a press tool.

[0209] In other words, the second bearing **136** is press-fitted (e.g., press-connected or press-fixed) to the intermediate flange **130**.

[0210] The leaf spring **150** is arranged on an opposite side of the intermediate flange **130** with

respect to the side, in which the second bearing **136** is pressed.

[0211] Turning to FIG. **9**, a part of the assembly method of FIG. **6** is schematically shown.

[0212] In particular, with reference to step **S30**, the first housing **126** being assembled with the electromagnet **108** of the brake device **100** and with the first bearing **134** is schematically shown.

[0213] In this respect, the electromagnet **108** is disposed in the first housing **126** and is fixed therein by means of at least one fastener **304**, for example, two fasteners **304**.

[0214] In other words, the electromagnet **108** is fixed to the first housing **126** by at least one fastener **304**.

[0215] The fastener **304** is configured as a screw fastener.

[0216] A spacer **306** is arranged in the electromagnet **108**, in particular in a passage for receiving the motor shaft **122**, which is formed within the electromagnet **108**.

[0217] The spacer **306** is pot shaped, i.e., the spacer **306** has a cylindrical hollow body with a collar at one end thereof, which collar protrudes outwards thereof.

[0218] The spacer **306** may be configured as a slide bearing.

[0219] As can be seen in FIG. **9**, the first bearing **134** is pressed into the first housing **126** by means of a third tool **404**, in particular, a press tool.

[0220] While pressing the first bearing **134** into the first housing **126**, the first housing **126** is countered into the pressing direction by a fourth tool **406**, in particular, a press tool.

[0221] In other words, the first bearing **134** is press-fitted (e.g., press-connected or press-fixed) to the first housing **126**.

[0222] The first bearing **126** and the electromagnet **108** are disposed on the first housing **126** such as to sandwich the collar of the spacer **306** between themselves.

[0223] In other words, the collar of the spacer **306** is disposed between the first bearing **126** and the electromagnet **108**, to thereby being fixed between them and, accordingly, to the first housing **126**.

[0224] After pressing the first bearing **134** into the first housing **126**, a plate **308** is fixed to the first housing **126** for reliably securing the seat of the first bearing **134** in the first housing **126** (see also FIGS. **10** to **14**).

[0225] In this respect, the plate **308** is disposed on the first housing **126** such as the first bearing **134** being sandwiched between the plate **308** and the first housing **126**. For example, the plate **308** is disposed partly on top of the first bearing **134**.

[0226] The plate **308** is fixed to the first housing **126** by means of at least one fastener, for example, two fasteners (not shown in FIG. **9**, but see FIGS. **10** to **14**).

[0227] Said fastener is configured as a screw fastener.

[0228] Turning to FIG. **10**, a part of the assembly method of FIG. **6** is schematically shown.

[0229] In particular, with reference to step **S30**, the first housing **126** being assembled with the motor shaft **122** is schematically shown. Especially, the first housing **126** being partly assembled as seen in FIG. **9** is further assembled with the motor shaft **122**.

[0230] In this context, the expression partly assembled first housing **126** may be understood that the first housing **126** may be assembled or provided or mounted with a part of a/the plurality of elements of the motor stack **110** as particularly described with respect to FIG. **2A**.

[0231] As can be seen in FIG. **10**, the motor shaft **122** is pressed into the (partly assembled) first housing **126** by means of a fifth tool **408**, in particular, a press tool.

[0232] Particularly, the motor shaft **122** is pressed into the first bearing **134** (i.e., the rotatable part of the first bearing **134**) by means of the fifth tool **408** and extends through the spacer **306**.

[0233] While pressing the motor shaft **122** into the (partly assembled) first housing **126**, i.e., the first bearing **134** and through the spacer **306**, the (partly assembled) first housing **126**, in particular, the spacer **306** of the (partly assembled) first housing **126**, is countered into the pressing direction by a sixth tool **410**, in particular, a press tool.

[0234] In other words, the motor shaft **122** is press-fitted (e.g., press-connected or press-fixed) to the first bearing **134** (i.e., the rotatable part of the first bearing **134**).

[0235] Turning to FIG. **11**, a part of the assembly method of FIG. **6** is schematically shown.

[0236] In particular, with reference to step **S30**, the first housing **126** being assembled with the motor shaft **122**, the intermediate flange sub assembly, and the rotatable part **140-1** of the driving assembly **140** is schematically shown.

[0237] As can be seen in FIG. **11**, the flange **138** together with the permanent magnet **106** (i.e., the intermediate flange sub assembly) is arranged on the motor shaft **122**.

[0238] In particular, the flange **138** together with the permanent magnet **106** is disposed adjacent (e.g., next) to the spacer **306** and, accordingly, adjacent to the electromagnet **108**.

[0239] On the motor shaft **122**, the flange **138** is supported by the spacer **306**.

[0240] In particular, the flange **138** is fixed to the motor shaft **122** by a/the above-described shaft hub connection **124**.

[0241] In this regard, assembling/building tolerances, e.g., very defined assembling/building tolerances, may be important in order to secure the above-mentioned gap **G** between the electromagnet **108** and the permanent magnet **106** for obtaining the hysteresis effect.

[0242] Further, the rotatable part **140-1** of the driving assembly **140** is arranged on the motor shaft **122**.

[0243] In particular, the rotatable part **140-1** of the driving assembly **140** is disposed adjacent (e.g., next) to the flange **138**.

[0244] On the motor shaft **122**, the rotatable part **140-1** of the driving assembly **140** is supported by the flange **138**.

[0245] Hence, on the motor shaft **122**, the flange **138** is disposed between the spacer **306** and the rotatable part **140-1** of the driving assembly **140**.

[0246] Adjacent (e.g., next) to the rotatable part **140-1** of the driving assembly **140**, fixing means for fixing the above-described elements (i.e., the flange **138** and the rotatable part **140-1** of the driving assembly **140**) on the motor shaft **122** are disposed.

[0247] Said fixing means are configured as a slotted nut **310** and a retaining washer **312**.

[0248] The retaining washer **312** is disposed adjacent (e.g., next) to the rotatable part **140-1** of the driving assembly **140**.

[0249] Hence, on the motor shaft **122**, the rotatable part **140-1** of the driving assembly **140** is disposed between the retaining washer **312** and the flange **138**.

[0250] The slotted nut **310** is disposed adjacent (e.g., next) to the retaining washer **312**.

[0251] In this respect, in a mounted state, the retaining washer **312** engages with the slotted nut **310** for fixing the slotted nut **310** in its mounted state on the motor shaft **122**.

[0252] In other words, the retaining washer **312** is configured to engage a/the slotted nut **310** for fixing the slotted nut **310**, e.g., for fixing the slotted nut **310** with respect to rotation.

[0253] The slotted nut **310** is screwed to a threaded portion of the motor shaft **122** by means of a seventh tool, in particular, a tool having an adapter for engaging a/the slotted nut **310**, particularly, a torque wrench tool being configured for engaging a/the slotted nut **310**.

[0254] Further, an eighth tool **414** is provided for mounting the slotted nut **310** to the motor shaft **122**.

[0255] The motor shaft **122** and, hence, the (partly assembled) first housing **126** are temporarily mounted to the eighth tool **414**, e.g., by means of fasteners, in particular, screw fasteners (for example two screw fasteners) for providing a fixation of the motor shaft **122** and, hence, the (partly assembled) first housing **126** during mounting the slotted nut **310**.

[0256] In other words, the motor shaft **122** is temporarily fixed to the eighth tool **414** with respect to rotation during (e.g., rotatably) assembling the slotted nut **310**.

[0257] Turning to FIGS. **12A** to **12C**, alternatives of a part of the assembly method of FIG. **6** are schematically shown, respectively.

[0258] In particular, FIGS. **12A** to **12C** respectively show different approaches of mounting the non-rotatable part **140-2** of the driving assembly **140** to the (partly assembled) first housing **126**.

[0259] As can be seen in FIGS. 12A to 12C, the non-rotatable part **140-2** of the driving assembly **140** is arranged in the (partly assembled) first housing **126**.

[0260] Particularly, the non-rotatable part **140-2** of the driving assembly **140** is correspondingly disposed with respect to the rotatable part **140-1** of the driving assembly **140**.

[0261] In this respect, the non-rotatable part **140-2** of the driving assembly **140** substantially surrounds the rotatable part **140-1** of the driving assembly **140**.

[0262] The non-rotatable part **140-2** of the driving assembly **140** is fixed to the first housing **126** as known in the art, for example, by means of a form fit, or press fit, etc.

[0263] Now turning to FIG. 12A, a first approach of said approaches is schematically shown.

[0264] A guide tool **416** for mounting the non-rotatable part **140-2** of the driving assembly **140** to the first housing **126** is provided.

[0265] The guide tool **416** comprises at least one guide rod **418** and at least one guide magnet **420**.

[0266] For example, the guide tool **416** comprises three guide rods **418** and three guide magnets **420**.

[0267] The guide rods **418** are evenly disposed around a mounting space for the first housing **126**, i.e., the guide rods **418** are disposed around a/the mounting space for the first housing **126** at 120 degrees with respect to each other.

[0268] Each guide magnet **420** is assigned to a respective guide rod **418**.

[0269] The respective guide magnet **420** is slidably attached to its assigned guide rod **418**.

[0270] Each guide magnet **420** is configured to attract the non-rotatable part **140-2** of the driving assembly **140** and to thereby hold the non-rotatable part **140-2** of the driving assembly **140** for mounting.

[0271] Hence, the (partly assembled) first housing **126** is disposed on the mounting space and the non-rotatable part **140-2** of the driving assembly **140** is manually clamped to the guide magnets **420** and mounted to the (partly assembled) first housing **126** in a sliding movement along said guide rods **418**.

[0272] Then, in its final mounting state, the non-rotatable part **140-2** of the driving assembly **140** is manually released from the guide tool **416**.

[0273] Now turning to FIG. 12B, a second approach of said approaches is schematically shown.

[0274] A guide tool **422** for mounting the non-rotatable part **140-2** of the driving assembly **140** to the first housing **126** is provided.

[0275] The guide tool **422** comprises at least one guide rod **424** and at least one clamping device **426**.

[0276] For example, the guide tool **422** comprises two guide rods **424** and one clamping device **426**.

[0277] The guide rods **424** are disposed on opposite sides of a mounting space for the first housing **126**, i.e., the guide rods **424** are disposed around a/the mounting space for the first housing **126** at 180 degrees with respect to each other and the mounting space.

[0278] The clamping device **426** is configured to surround the non-rotatable part **140-2** of the driving assembly **140** and to be reversibly arrangeable between a clamping state, in which the non-rotatable part **140-2** is clampable, and a releasing state, in which the non-rotatable part **140-2** is releasable.

[0279] For example, the clamping device **426** may be configured as a clamping ring or a clamping collar.

[0280] The clamping device **426** is fixed to the guide rods **424** and is able to move along the guide rods **424**.

[0281] For mounting, the (partly assembled) first housing **126** is disposed on the mounting space and fixed to the guide tool **422** by means of fasteners, e.g., screw fasteners.

[0282] The non-rotatable part **140-2** of the driving assembly **140** is clamped in the clamping device **426** and (subsequently) mounted to the (partly assembled) first housing **126** in a sliding movement

along said guide rods **424**.

[0283] Then, in its final mounting state, the non-rotatable part **140-2** of the driving assembly **140** is released from the guide tool **422**.

[0284] In the present embodiment, the clamping device **426** may be guided between the guide rods **424** (e.g., by means of a form fit) and/or may be guided over slide bushings and/or sleeve bearings arranged in the clamping device **426** for sliding along the guide rods **424**.

[0285] Now turning to FIG. **12C**, a third approach of said approaches is schematically shown.

[0286] A guide tool **428** for mounting the first housing **126** to the non-rotatable part **140-2** of the driving assembly **140** is provided.

[0287] The guide tool **428** comprises at least one guide rod **430** and at least one clamping device **432**.

[0288] For example, the guide tool **422** comprises one guide rods **430** and one clamping device **432**.

[0289] The guide rod **430** is disposed in a center of a mounting space for the non-rotatable part **140-2** of the driving assembly **140**.

[0290] The clamping device **432** is configured to surround the non-rotatable part **140-2** of the driving assembly **140** and to be reversibly arrangeable between a clamping state, in which the non-rotatable part **140-2** is clampable, and a releasing state, in which the non-rotatable part **140-2** is releasable.

[0291] For example, the clamping device **432** may be configured as a clamping ring or a clamping collar.

[0292] For mounting, the non-rotatable part **140-2** is disposed on the mounting space and fixed thereto by means of the clamping device **432**.

[0293] Then, the (partly assembled) first housing **126** is engaged to the guide rod **430** by means of the motor shaft **122**, which is configured as a hollow shaft, by receiving the guide rod **430** in the motor shaft **122**.

[0294] Subsequently, the (partly assembled) first housing **126** is assembled to the non-rotatable part **140-2** in a sliding movement along said guide rod **430**.

[0295] Then, in its final mounting state, the (partly assembled) first housing **126** together with the non-rotatable part **140-2** of the driving assembly **140** is released from the guide tool **428**.

[0296] Alternatively, it may also conceivable that the guide tool **428** comprises at least one guide rod being disposed on a side of the mounting place and at least one clamping device for clamping the (partly assembled) first housing **126** in a surrounding manner, wherein the clamping device is movable along the guide rod for mounting the (partly assembled) first housing **126** to the non-rotatable part **140-2** of the driving assembly **140**.

[0297] Turning to FIG. **13**, a part of the assembly method of FIG. **6** is schematically shown.

[0298] In particular, with reference to step **S40**, the second housing **128** is mounted to the first housing **126**, i.e., to the first housing sub assembly.

[0299] For example, the second housing **128** is screwed to the first housing **126**, e.g., by means of a screw fastener.

[0300] Turning to FIG. **14**, a part of the assembly method of FIG. **6** is schematically shown.

[0301] In particular, with reference to step **S50**, the intermediate flange sub assembly is mounted to the second housing **128** and to the first housing sub assembly having the second housing **128** mounted thereto.

[0302] As can be seen in FIG. **14**, the intermediate flange sub assembly is pressed onto the motor shaft **122** by means of a ninth tool **434**, in particular, a press tool.

[0303] In particular, the second bearing **136**, i.e., the rotatable part of the second bearing **136**, is pressed onto the motor shaft **122** by means of a ninth tool **434**, in particular, a press tool.

[0304] While pressing the second bearing **136** onto the motor shaft **122**, the motor shaft **122** is countered into the pressing direction by a tenth tool **436**, in particular, a press tool.

[0305] In other words, the second bearing **136** is press-fitted (e.g., press-connected or press-fixed) to the motor shaft **122**.

REFERENCES

[0306] **10** joint arrangement [0307] **20** robot arm [0308] **100** brake device [0309] **102** stator assembly of brake device [0310] **104** rotor assembly of brake device [0311] **106** permanent magnet [0312] **108** electromagnet [0313] **110** motor stack [0314] **112** pulley [0315] **114** belt [0316] **116** control unit [0317] **118** rotor assembly of motor stack [0318] **120** stator assembly of motor stack [0319] **122** central motor shaft [0320] **124** shaft hub connection [0321] **126** first housing [0322] **128** second housing [0323] **130** intermediate flange [0324] **132** cover [0325] **134** first bearing [0326] **136** second bearing [0327] **138** flange [0328] **140** driving assembly [0329] **142** drive PCB [0330] **144** PCB holding structure [0331] **146** encoder [0332] **148** pretension spring [0333] **150** leaf spring [0334] **200** sub arm [0335] **202** further sub arm [0336] **204** adapter assembly [0337] **206** surgical instrument [0338] **300** brake armature [0339] **302** fastener [0340] **304** fastener [0341] **306** spacer [0342] **308** plate [0343] **310** slotted nut [0344] **312** retaining washer [0345] **400** first tool [0346] **402** second tool [0347] **404** third tool [0348] **406** fourth tool [0349] **408** fifth tool [0350] **410** sixth tool [0351] **412** seventh tool [0352] **414** eighth tool [0353] **416** guide tool [0354] **418** guide rod [0355] **420** guide magnet [0356] **422** guide tool [0357] **424** guide rod [0358] **426** clamping device [0359] **428** guide tool [0360] **430** guide rod [0361] **432** clamping device [0362] **434** ninth tool [0363] **436** tenth tool [0364] **S1 to S4** steps of method [0365] **S10 to S60** steps of assembly method [0366] **G** gap

Claims

1. Joint arrangement for a robot arm, especially a robot arm for use in microsurgery, comprising: at least one brake device being configured for engagement and for disengagement and comprising a stator assembly and a rotor assembly, wherein the brake device is further configured to be operable at a minimal power consumption in a disengagement state.
2. The joint arrangement of claim 1, wherein the brake device comprises: at least one permanent magnet for providing a braking effect during engagement and being assigned to the rotor assembly, and at least one electromagnet for releasing the braking effect during disengagement and being assigned to the stator assembly, wherein the permanent magnet and the electromagnet are arranged or arrangeable to each other with a gap between themselves such as to allow the brake device to be operable at a minimal power consumption.
3. The joint arrangement of claim 2, wherein the gap is smaller than $1\text{ mm}\pm 10\%$.
4. The joint arrangement of claim 2, wherein the joint arrangement further comprises at least one motor stack having a rotor assembly and a stator assembly, wherein the brake device is arranged in connection with or within the motor stack.
5. The joint arrangement of claim 4, wherein the rotor assembly of the brake device is configured as a part of the rotor assembly of the motor stack, and the stator assembly of the brake device is configured as a part of the stator assembly of the motor stack.
6. The joint arrangement of claim 4, wherein the motor stack comprises at least one spring being configured to provide a spring force to the brake device .
7. Robot arm comprising: a joint arrangement of claim 1.
8. Method for operating a brake device in a joint arrangement for a robot arm the method at least comprising the following step: the brake device is provided and configured for engagement and for disengagement, the brake device comprising a stator assembly and a rotor assembly, wherein the brake device is further operated at a minimal power consumption in a disengagement state.
9. The method of claim 8, wherein the brake device is further configured for engagement at an engagement threshold value and for disengagement at a disengagement threshold value being higher than the engagement threshold value, and the method further comprises the following steps:

providing full power to the brake device by applying Pulse Width Modulation for disengaging the brake device, reducing the power provided to the brake device to a first predetermined power value by applying PWM, wherein the first predetermined power value is set between the disengagement threshold value and the engagement threshold value, and maintaining the power provided to the brake device at the first predetermined value by applying PWM for keeping the brake device in disengagement.

10. The method of claim 9, wherein the method further comprises reducing the power provided to the brake device to a second predetermined power value by applying PWM for engaging the brake device, wherein the second predetermined power value is set below the engagement threshold value.

11. The method of claim 9, wherein the first predetermined power value is half of the full power being providable or provided to the brake device by applying PWM for disengaging the brake device.

12. The method of claim 9, wherein the full power being providable or provided to the brake device by applying PWM for disengaging the brake device includes a 24V signal.

13. The method of claim 9, wherein the disengagement threshold value is $\frac{5}{8}$ of the full power.

14. The method of claim 9, wherein the engagement threshold value is $\frac{1}{3}$ of the full power.

15. The method of claim 9, wherein providing full power to the brake device by applying PWM for disengaging the brake device is maintained for a first predetermined time period, and maintaining the power provided to the brake device at the first predetermined power value by applying PWM for keeping the brake device in disengagement is maintained for a second predetermined time period, wherein the second predetermined time period is longer than the first predetermined time period.

16. The method of claim 8, wherein the brake device comprises at least one permanent magnet for providing a braking effect during engagement and at least one electromagnet for releasing the braking effect during disengagement.

17. The method of claim 8, wherein the brake device is arranged in connection with or within a motor stack of the joint arrangement, optionally, wherein the motor stack comprises at least one spring being configured to provide a spring force to the brake device.
